FAPS: Failure-Aware Policy Shielding for Vision-Language-Action Models

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Problem Statement & Motivation

VLA models generalize across manipulation tasks but remain brittle under distribution shift. Sensor noise, viewpoint drift, and actuation jitter corrupt hidden states; downstream denoising passes compound the error — the policy commits to failure with no detect-or-correct mechanism.

FAPS is the first *triggered*, *learned*, safety-metriced activation-space shield: fires only when calibrated risk exceeds a threshold, injects a flow-model correction conditioned on failure type, falls back safely when inactive — no weights modified.

Prior Work & Positioning

Activation steering — RepE, ITI, ROME, MEMIT steer LLMs via additive residuals or null-space edits. None address continuous-action VLAs.

VLA robustness — VLA-OS, Don't Blind the VLA, LAE, Cerebellar, ReSteer: layer analysis, activation intervention, action-space correction, steerability. None are triggered or calibrated.

Concurrent — COAST: always-on closed-form conceptor steering ($h' = hM$). FAPS adds triggering, a learned flow model, and adversarial eval.

System Architecture

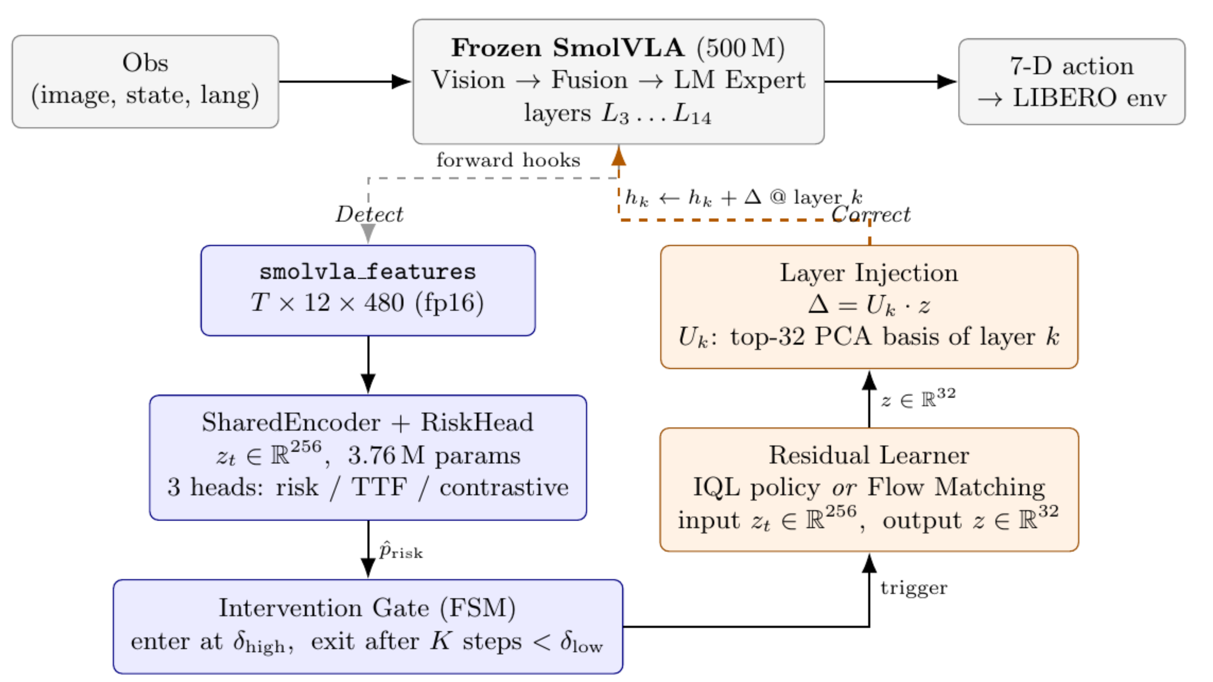

Figure 1. FAPS pipeline (inference-time). Hooks at SmolVLA layers $\{4, 5, 6, 11, 12, 13\}$ feed the risk head ($\hat{p}_{\text{risk}}$, z_t); the hysteresis FSM gates injection; flow model adds $\Delta = Uz$ at layer k. No VLA weights modified.

Data & Perturbations

5,117 labeled rollouts of SmolVLA on LIBERO across 11 perturbation types. Labels: `fail_in_H` ($H = 30$) and failure-type tag (collision, drop, grasp misalignment, grasp precision).

- **Eval only:** joint-angle noise (hard negatives)
- **Train+eval:** action noise, viewpoint shift, Gaussian blur, windowed variants (×2 each)
- **Trigger retrain:** 999 windowed perturbations + 250 nominal successes; timeout nominal failures excluded

Risk Head

SharedEncoder (3.76M): temporal CNN + cross-layer attention over 6 layers → $\hat{p}_{\text{risk}}$ and $z_t \in \mathbb{R}^{256}$.

$\mathcal{L} = \lambda_{\text{BCE}}\mathcal{L}_{\text{BCE}} + \lambda_{\text{TTF}}\mathcal{L}_{\text{TTF}} + \lambda_{\text{NT-Xent}}\mathcal{L}_{\text{NT-Xent}}$

BCE supervises risk; TTF adds time-to-failure granularity; NT-Xent induces failure-type geometry in z_t (silhouette $+0.168$).

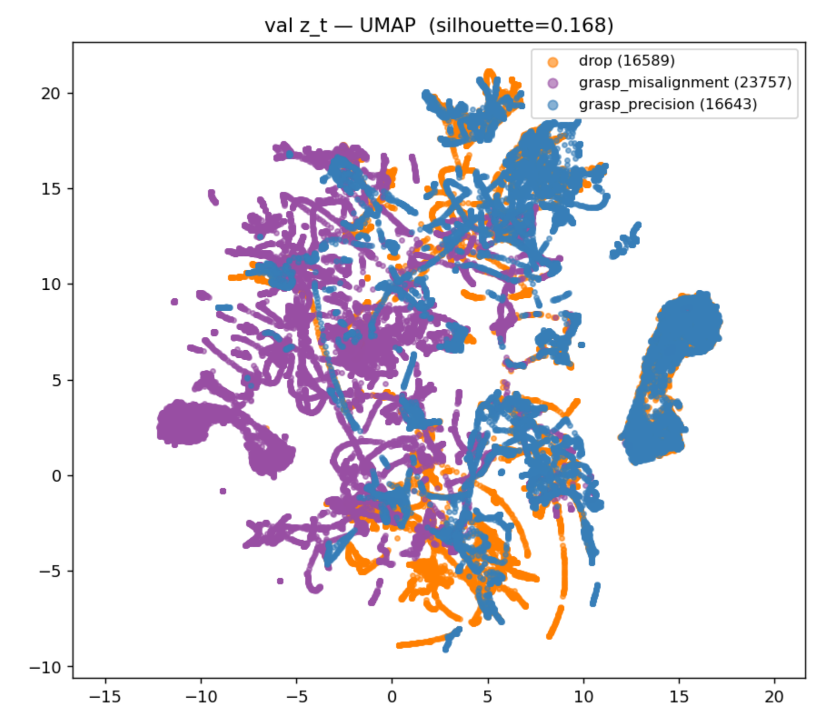

Figure 2. UMAP of z_t (Run C). Failure types cluster clearly (silhouette $+0.168$).

Approach: Flow-Matching Residual & FSM

Flow model conditioned on z_t generates $z \in \mathbb{R}^{48}$; projected via top-48 PCA basis U_k ($\sim$95% variance): $\Delta = U_k z$, $h'_k = h_k + \Delta$.

Layer candidates: C12/C13 conceptor+failure flow runs produce substantial residuals (loss ~ 0.30, $\|\Delta\| \sim 1.5$, 93% nonzero). Layer-11 closed-loop eval below isolates the trigger failure mode before retrain.

Hysteresis FSM: enter at $\hat{p} \geq 0.7$; exit after $K = 5$ steps below 0.4; any fault → unshielded fallback.

Risk Head Ablation

Table 1. Loss ablation. BCE detects risk; NT-Xent adds geometry.

Run	wAUC	Silh.	ECE	BSkill
A (BCE only)	0.928	−0.046	0.088	−0.067
B (BCE+TTF)	0.927	−0.040	0.093	−0.12
C (BCE+TTF+NT-Xent)	0.917	+0.168	0.077	+0.018
D (NT-Xent only)	0.440	+0.125	—	—

Run C selected. NT-Xent costs -0.010 wAUC vs. B, raises silhouette by $+0.208$, improves calibration (ECE 0.093→0.077).

Layer Probe

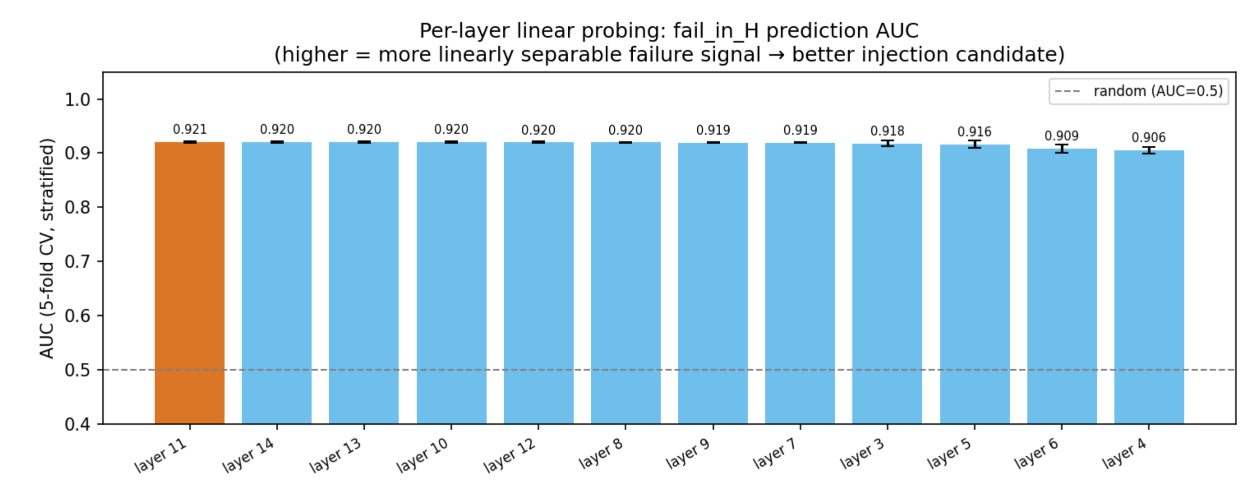

Figure 3. Linear probe AUC per action-expert layer (≈ 0.92 flat across all 12). Failure is linearly separable everywhere in SmolVLA's activations. Layer 11 (AUC 0.921) selected: mid-depth lets downstream layers regenerate coherent actions from the corrected state. Final layer comparison via closed-loop SR delta is in progress.

Experimental Findings: Shield Evaluation

Latest demo eval: B6 layer-11 conceptor-flow residual with the retrained Run C risk head. This is demo-scale, but now includes online shielded vs. unshielded evidence on action-noise tasks.

Table 2. B6 online demos. FAPS improves task-level action-noise success, but trigger false positives remain the limiting factor.

Eval	N	Base SR	FAPS SR	ΔSR	FPR
action focus T0–4	50	—	0.74	—	0.86
action T3	10	0.30	0.60	+0.30	0.91
action T4	10	0.80	0.90	+0.10	0.66
strong all-pert T0–2	90	pend.	0.789	pend.	0.57
medium all-pert T0–2	90	pend.	0.789	pend.	0.68

Video check: success clips place the object near the plate; the failure clip runs to timeout without placement. The residual can help, but the gate still fires too often.

Insights & Takeaways

Risk is accessible in VLA activations. BCE+TTF reaches wAUC 0.927; adding contrastive supervision preserves detector quality while making the representation useful for correction.

Contrastive geometry is the key design choice. NT-Xent raises silhouette $-0.040 \to +0.168$ at only -0.010 wAUC cost — enabling the flow model to condition on failure type.

The current online bottleneck is trigger precision. B6 layer-11 demos improve action-noise task success, but trigger on nearly every episode. Threshold and persistence tuning are the next lever.

Limitations & Future Work

- Simulation only (LIBERO); real-world not validated.
- Single VLA (SmolVLA 500M).
- Closed-loop demos are positive but still small-scale.
- Full all-perturbation unshielded baseline is pending.
- Trigger calibration lowers false alarms offline, but online FPR is still too high.
- PCA basis U_k vs. null-space basis U_{null} and action-sensitive U_{potent} not yet compared.
- Future extensions: failure-type routing, multi-task VLAs, real-world transfer, and online adaptation.

Planned Remaining Work

- Complete all-perturbation unshielded baselines for tasks 0–4.
- Sweep $\delta_{\text{high}} \in [0.92, 0.97]$ and $K \in \{5, 7\}$ to reduce false positives.
- Run closed-loop C sweep; pick best layer by success-rate delta and intervention precision.
- Run E ablations: U_k vs. U_{potent} vs. U_{null} and first/last/all injection timing.
- Tune trigger threshold and injection strength, then report final shielded vs. unshielded success rates.

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